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Borrowed plant defences: Deterring browsers using a forestry by-product

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ABSTRACT

Mammalian browsers can cause ecologically and economically significant damage to important tree species, particularly at the vulnerable seedling stage. Some tree species or particular parts of trees are less preferred as a result of chemical defences – harnessing such defences could reduce the appeal of more preferred trees and thus their loss to browsing. We tested the potential of birch bark extract, a forestry by-product, to protect seedlings of Scots pine *Pinus sylvestris* from two globally significant browsers: moose *Alces alces* and red deer *Cervus elaphus*. We quantified the responses by captive animals to extract application as a function of both the distance between seedlings and whether the extract was applied to all or alternate seedlings. Both moose and red deer consumed pine seedlings less frequently when seedlings were treated with birch bark extract, but their responses differed when only alternate seedlings were treated. At inter-seedling distances equivalent to those used in forestry, red deer browsed untreated seedlings more frequently when alternate seedlings were treated than when none were treated (associational susceptibility), whereas moose browsed untreated seedlings less frequently with treated than untreated neighbours (associational refuge). These neighbourhood effects were not evident at inter-seedling distances three times larger or smaller than those used in forestry. There was also no significant difference in the frequency of browsing on treated seedlings in alternate compared to completely treated arrangements at any inter-seedling distance. Red deer removed significantly less foliage from treated than untreated seedlings once browsed. Browsing by moose followed a similar but non-significant trend. The relative number of untreated to treated seedlings browsed was 2:1 for red deer and 4:1 for moose. Red deer used a finer scale of selection than moose for choosing between seedlings, consistent with their smaller body mass. However, the coarser-scale of selection used by moose resulted in consumption of more untreated relative to treated seedlings. Our study highlights the potential of existing plant defensive chemistry, harvested from forestry by-products, to reduce herbivore damage to tree seedlings. We also demonstrate the need to consider both plant associational effects and the foraging strategies of the dominant herbivores in applying repellents.

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1. Introduction

Browsing of trees, particularly at the vulnerable juvenile stage, has significant effects on plant community structure and composition in many parts of the world (Côté et al., 2004; Moles and Westoby, 2004). Not only does it influence the natural regeneration of trees in forests, with cascading effects for ecosystems, it can hamper revegetation efforts and damage stocks in forestry plantations (Bulinski and McArthur, 1999; Meers and Adams, 2003; Nuttle et al., 2011). Although exclusion fencing and hunting are sometimes employed to manage browsing by herbivorous mammals, such practices are often undesirable because they can be difficult to implement (e.g. due to terrain and safety concerns), prohibitively expensive over large areas, interfere with conservation measures, elicit negative public perceptions, and conflict with the needs of game hunters (Mattsson, 1990; Di Stefano, 2004; Miller et al., 2009). One alternative is to manipulate the foraging choices of herbivores by using repellents to reduce the appeal of otherwise preferred plants (Nolte, 1999).

Many commercial repellents exist, only some of which have been tested experimentally, with variable results (e.g. Harris et al., 1983; Arnould and Signoret, 1993; Woolhouse and Morgan, 1995; El Hani and Conover, 1998; Nolte, 1998). Some repellents have shown promise as browsing deterrents but they often require valuable raw materials to produce, involve introduction of novel

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materials into an environment with unknown secondary effects, and reduce in effectiveness over time (Engle and Shimmel, 1984; Murray et al., 2006). An opportunity that has been largely overlooked is to take advantage of existing plant defensive chemistry, particularly from forestry by-products. Over their long history of co-existence with herbivores, plants have evolved a gamut of physical and chemical characteristics that deter consumption (Ehrlich and Raven, 1964). The latter include compounds that are either toxins (e.g. terpenes) or digestibility reducers (e.g. tannins, phenolic compounds) (Belovsky and Schmitz, 1994; Foley and McArthur, 1994). These defences result in some plants or parts of plants being less preferred or even avoided by herbivores either by conferring some negative consequences on handling and/or consumption, or by being associated with cues indicating such consequences (Provenza et al., 1992). It is therefore plausible that harnessing these chemical plant defences could help defend preferred plants from consumption.

We tested the potential of one such deterrent, birch bark extract (*Betula pubescens* and *B. pendula*), to protect commercially valuable Scots pine seedlings (*Pinus sylvestris*) from browsing by two large ungulates, red deer *Cervus elaphus* and European moose *Alces alces*. Birch bark contains tannins, phenols and terpenes (Palo, 1984; Vainiotalo et al., 1991), with several compounds implicated in reducing digestibility (Palo, 1985; Sunnerheim et al., 1988). The addition of birch bark extract to food pellets reduced consumption by another ungulate, the fallow deer (*Dama dama*; Bergvall et al., 2013). We extended this study to test the extract on a commercially important tree and its susceptibility to browsing by two globally significant ungulates.

In many areas of Scandinavia, red deer and moose co-exist and are major contributors to the mammalian browsing pressure on conifers (Nichols and Spong, 2014). Intense browsing by these herbivores has affected other components of ecosystems (Pastor et al., 1993; Pedersen et al., 2007) and reduced the profitability of the forestry industry (Heikkilä and Härkönen, 1996). Deterring red deer and moose from browsing Scots pine is particularly important as even very low levels of damage significantly reduce its production value (Nilsson et al., 2016).

Our study moved beyond the traditional tests of repellent effectiveness by considering both the distance between plants and the plant neighbourhood created by the application scheme. Foraging by large herbivores involves decisions made at a hierarchy of spatial scales from landscapes to parts of plants (Senft et al., 1987; Kotliar and Wiens, 1990); to understand how a repellent operates, it is therefore critical to consider the spatial context in which it is applied. We tested the effectiveness of birch bark extract at inter-seedling distances equivalent to those used in commercial forestry operations, and one larger and one smaller distance. There is a cost to foraging ungulates of greater distances between food items, manifested in faster movement between patches, larger and increased number of bites, and more time spent in patches (Shipley and Spalinger, 1995). These behaviours likely account for the pattern of moose browsing observed in Sweden, with greater damage at lower densities of Scots pine (Andrén and Angelstam, 1993). However, the plant neighbourhood can also affect herbivory, either by creating refuge for neighbours or by increasing their susceptibility to herbivory (Tahvanainen and Root, 1972; Atsatt and O'Dowd, 1976). Associational plant effects have been demonstrated across many taxa including large mammalian herbivores (Callaway et al., 2005; Bergvall et al., 2008; Stutz et al., 2015). For herbivores with movement capacities comparable to deer, the magnitude of associational effects decreases with greater distance between plants (Champagne et al., 2016); we therefore explored these effects as a function of inter-seedling distance.

Associational effects are influenced not only by relative plant palatability (based on nutritional content and plant defences), but also by the scale at which herbivores predominantly make foraging decisions (Tahvanainen and Root, 1972; Atsatt and O'Dowd, 1976). By applying a repellent to a plant, the plant's appeal to the herbivore should be reduced. Repellent-treated plants may protect their untreated neighbours from herbivory by reducing the appeal of the plant patch. This outcome is contingent on the herbivore making decisions based primarily on the palatability of the whole patch relative to other patches, rather than between plants within a patch (i.e. the between-patch scale). This is the classic case of associational refuge via the repellent plant mechanism (Atsatt and O'Dowd, 1976). In this way, not all seedlings may need to be treated to be effectively protected from herbivory, substantially reducing the amount of repellent needed in commercial operations. However, the inverse effect may be observed if herbivores choose primarily between plants within a patch (i.e. the between-plant scale). Untreated plants may then be more vulnerable to herbivores because they are of higher palatability relative to their treated neighbours, leading to associational susceptibility via the neighbour contrast mechanism (Bergvall et al., 2006). Being in a mixed palatability neighbourhood can also have consequences for repellent-treated plants: between-patch selection would result in greater consumption of treated plants due to the presence of the more palatable untreated plants (susceptibility via shared doom; Wahl and Hay, 1995), whereas between-plant selection would result in reduced consumption due to lower relative palatability (refuge via attractant-decoy or neighbour contrast; Atsatt and O'Dowd, 1976; Bergvall et al., 2006). In this study, we explored the potential for any associational plant effects that could result from different inter-seedling distances and repellent application schemes.

The effectiveness of a chemical plant defence may vary by herbivore species: they may differ in the physiological and behavioural mechanisms they have evolved for dealing with plant secondary compounds, and exhibit differing foraging strategies based on the quantity and quality of food they require for self-maintenance. Moose are both larger than red deer and consume a greater proportion of browse plants, such as trees and shrubs, as part of their diet (Hofmann, 1989). This may influence the relative foraging behaviours of these two ungulates, and we thus we expected that they could respond differently to seedlings in different neighbourhoods and at different inter-seedling distances. For both red deer and moose, we tested whether: (1) birch bark extract reduced the probability that a pine seedling was browsed; (2) inter-seedling distance affected browse probability overall; and (3) associational plant effects existed at any of the inter-seedling distances. We therefore not only determined whether birch bark extract could deter browsing of pines by red deer and moose, but tested associational plant effect theory in an applied context, with implications for repellent application in multi-ungulate systems.

2. Material and methods

2.1. Herbivores

We conducted our experiments with captive red deer and moose held in large enclosures (12 ha and 6 ha, respectively) in Södermanland, south-central Sweden. The red deer herd was kept at Tovetorp Research Station (Zoology Department, Stockholm University) and consisted of 11 individuals including one adult male, five adult females, and their five offspring from the previous season (all male). The moose were located approximately 1 km away at the Öster Malma Hunting Association's Wildlife Park; the group included one adult male and one adult female, with

two juveniles (one male, one female) also sometimes in the same enclosure. The experiments were set-up within the enclosures where the animals lived and received their complete diet. Both red deer and moose had access to water, salt and minerals, and were fed a diet of pellets (Renfor SP, Lantmännen, Nyköping, Sweden; content: corn, milling and sugar beet by-products, minerals, vitamins, fat and vegetable oils) and silage; the red deer received these separately but they were given in a mixture for the moose. We tested animals over the winter in early 2016 and enclosures were covered in snow for most of this period; the availability of foliage was therefore limited to two evergreen tree species, Scots pine *Pinus sylvestris* and Norway spruce *Picea abies*, and they had access to woody tissue from deciduous trees and shrubs including silver birch *Betula pendula*, Norway maple *Acer platanoides*, aspen *Populus tremula*, rowan *Sorbus aucuparia*, alder *Alnus glutinosa*, hazel *Corylus avellana*, oak *Quercus robur*, and European raspberry *Rubus idaeus*. Red deer experiments were performed with permission from the Swedish National Board for Laboratory Animals (Dnr 14–14). Moose experiments were performed in accordance with the Swedish Board of Agriculture regulations ch.2, §19, 2012:26, L150.

2.2. Birch bark extract

We used an ethanol extraction technique described in Bergvall et al. (2013) to obtain the chemical compounds from approximately 200 kg of birch bark (*Betula pendula* and *B. pubescens*). The extraction was then dried at ambient temperature in a fume hood. The resulting powder was mixed at 100 g L^{-1} with a starch glue solution used in commercial plantations and nurseries for applying repellents to plants (75% tap water, 25% Flexcoat, Lyckeby Coatings AB).

2.3. Seedlings

We used *P. sylvestris* seedlings germinated in the nursery and planted outdoors for 2 years ('TePlus', Skogsplantor, Vibytorps nursery, south-central Sweden). The seedlings were stored frozen in cardboard boxes from the previous summer (mean height = 224 ± 8 mm, n = 30); this is a common method used for storing seedlings of freeze-tolerant species that are dormant over winter. To treat seedlings, we immersed them in the extract solution and hung them from a suspended line to dry. We used this method because the solution contained small sticky particles that blocked the nozzles of our hand-pressurized spray bottles, but we do not anticipate that this would be a problem for commercial sprayers. Each treated seedling required approximately 15 mL of extract solution to coat completely. The application of the extract solution gave the plants a slightly brown appearance, but this does not seem to have any detrimental effects on photosynthetic activity: 'TePlus' seedlings planted in an ungulate-exclusion area at Tovetorp Research Station grew a similar amount between June and November 2016 regardless of treatment (untreated = 11 ± 3 mm, starch glue only = 11 ± 2 mm, extract in starch glue solution = 15 ± 4 mm, n = 15 per treatment).

2.4. Field trials

To test the effect of repellent application at different inter-seedling distances, we constructed three circles in each enclosure, varying in diameter by a factor of three (3 m, 9 m, 27 m; Fig. 1a). With 16 seedlings per circle, this resulted in distances of 0.59 m, 1.76 m and 5.27 m to the nearest seedling in the small, medium and large circles, respectively. At each inter-seedling distance, we tested three extract application schemes: (1) none applied, (2) applied to alternate seedlings, and (3) applied to all seedlings.

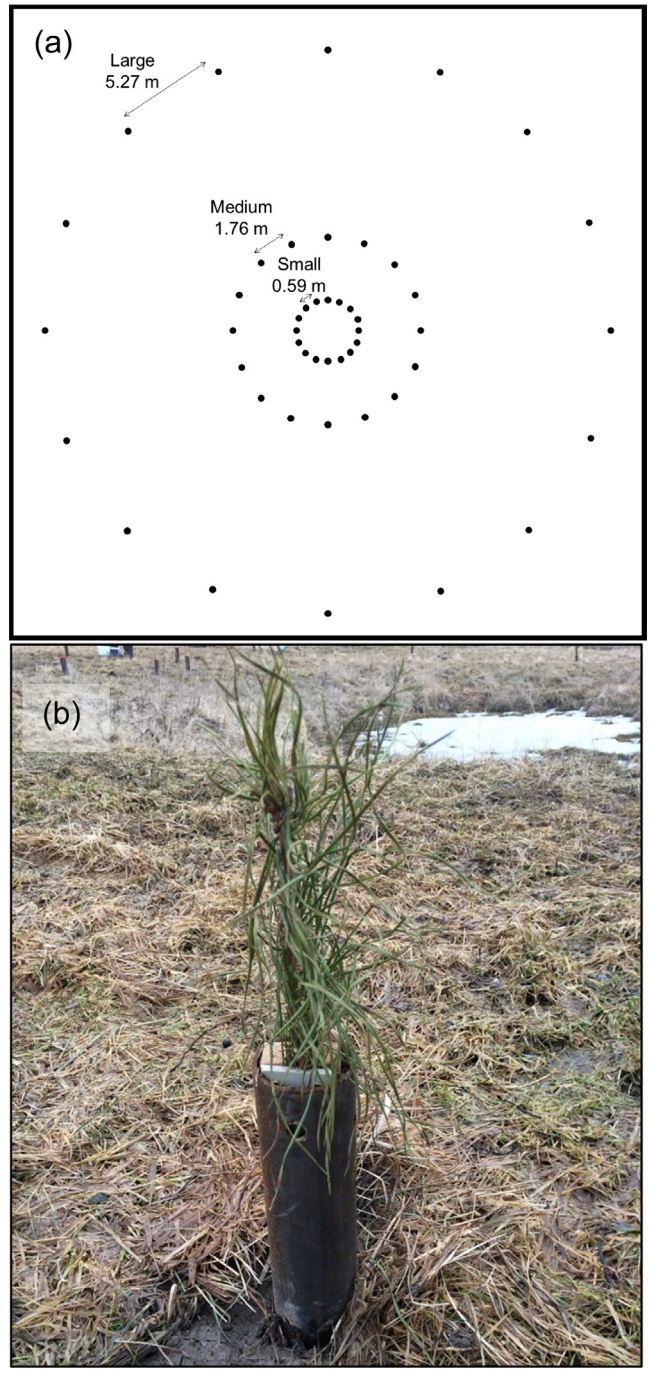

Fig. 1. (a) Layout of seedling holders used in the field. Only one inter-seedling distance (circle of holders) was tested in each trial. (b) Scots pine seedling (*Pinus sylvestris*) attached in holder for testing.

We therefore had nine treatments: three inter-seedling distances by three extract application schemes. We tested one treatment per trial, with each trial lasting 24 h. To facilitate the frequent change-over of treatments, we buried 16 hollow metal posts (500 mm × 60 mm in diameter, with the bases pinched closed) in the ground in each of three concentric circles (of the abovementioned diameters; Fig. 1b). We attached seedlings inside the posts using custom-made holders consisting of two pieces of grey plastic sheeting (30 mm × 60 mm × 5 mm) held together with four screws. Each post had a hole in the side at 30 mm from the rim that allowed the seedling holder to be firmly screwed in using a

handheld drill. To avoid potential injury to the animals, we placed plastic caps on the 32 posts not involved in the trial running on a given day. We tested the treatments in random order.

We conducted 36 trials (four replicates of each of the nine treatments) over 59 days from January through March 2016. We placed seedlings in their holders between 10:00–12:00 and returned at the same time the following day to record browsing damage and change the treatment. Browsing damage was recorded in terms of foliage lost on a sliding scale commonly used to assess tree seedling damage: 0%, 1–5%, 6–25%, 26–50%, 51–75%, 76–95%, 96–100% (Pietrzykowski et al., 2003; Miller et al., 2006). The distribution of ranges is based on the premise that estimation is more accurate at the extreme ends of the scale. Mid-points of these ranges were used for analyses. We assessed the presence of animals during each trial using motion-triggered infra-red cameras (Scoutguard SG550V).

2.5. Statistical analyses

To test the effect of extract application scheme (none, alternate, all) and inter-seedling distance (large, medium, small) on consumption, we initially used the percentage of foliage removed from seedlings as the response variable and trial as a random factor. However, the distribution of residuals was not normal and transformations failed to improve this. Given the high number of zeroes and values at the other extreme end of the scale, we reassessed the response as a binomial variable (0 = not browsed, 1 = browsed). We used the glmer procedure with the binomial distribution and logit link function, and specified the Bound Optimization by Quadratic Approximation algorithm (BOBYQA; Powell, 2009) to solve the model. We constructed an orthogonal contrasts matrix (weighted for sample size) to conduct the following planned comparisons: (1) treated versus untreated seedlings; (2) large and small versus medium inter-seedling distances; (3) large versus small inter-seedling distances; and at each scale, (4) untreated seedlings in alternate versus none schemes, and (5) treated seedlings in alternate versus all schemes. To form the contrasts, we constructed an interaction factor with 12 levels, corresponding to combinations of inter-seedling distance (three levels), extract application scheme (three levels), and whether or not the seedling was treated (two levels in 'alternate' application schemes but only one level in 'all' or 'none' schemes); for example, 'large:alternate: untreated'.

We used browsed seedlings only to test the effect of repellent application on browsing severity (i.e. how much foliage was removed from a seedling once browsed). A linear mixed effects models (lmer) was applied with percentage of foliage removed as the response, extract application (untreated/treated) as a fixed effect and trial as a random effect (specifying BOBYQA).

We performed all statistical analyses using R3.2.2 (R Core Team, 2016). We present all means with associated standard errors throughout.

3. Results

Video recordings and the consumption of pellets at each site showed that animals were present during every trial, with multiple individuals interacting with the seedlings presented. Red deer consumed some seedlings in 34 of the 36 trials; they did not browse any seedlings during one trial each of the 'alternate' and 'all' extract application schemes. Moose were more variable in their response, consuming seedlings in 22 of the 36 trials; they did not browse any seedlings in three 'none', four 'alternate' and seven 'all' trials.

3.1. Browsing probability

Red deer were more likely to browse untreated (0.38 ± 0.03) than treated seedlings (0.19 ± 0.02; Table 1; Fig. 2a). The probability of browsing a seedling was greater at medium inter-seedling distances (0.36 ± 0.03) than at the other distances (0.24 ± 0.03); the large and small inter-seedling distances did not differ significantly from each other (large = 0.26 ± 0.03, small = 0.22 ± 0.03). At medium inter-seedling distances, red deer were more than twice as likely to browse untreated seedlings when they had treated neighbours as when none of the seedlings were treated. This pattern did not occur at the other two inter-seedling distances. The probability of red deer browsing treated seedlings did not differ significantly whether extract was applied to alternate or all seedlings at any scale.

Consistent with the red deer, moose were also more likely to browse untreated (0.33 ± 0.03) than treated seedlings (0.08 ± 0.02; Table 2; Fig. 2b). The overall probability of browsing a seedling did not differ between inter-seedling distances. At medium inter-seedling distances, moose were almost six times less likely to browse untreated seedlings when they had treated neighbours than when none of the seedlings were treated. This pattern did not occur at the other two inter-seedling distances. The probability that moose browsed treated seedlings did not differ significantly whether extract was applied to alternate or all seedlings at any scale.

3.2. Browsing severity

When red deer browsed seedlings, they removed a significantly greater percentage of foliage from untreated than treated seedlings ($X_1^2 = 6.61$, $P = 0.010$; Fig. 3a). Browsing severity by moose followed a similar but non-significant trend, with greater variance in foliage removed from treated seedlings ($X_1^2 = 3.44$, $P = 0.064$; Fig. 3b). Both herbivores browsed a greater number of untreated than treated seedlings, by a factor of two by red deer and four by moose (Fig. 3a and b).

4. Discussion

We showed that birch bark extract, derived from a forestry by-product, significantly reduced the probability of pine seedling browsing by both red deer and moose. This supports the results of previous research where fallow deer reduced intake of food pellets that were treated with birch bark extract (Bergvall et al., 2013); we therefore suggest that the extract is unpalatable to a large proportion of ungulates resident in Scandinavian forests and may be a viable option for reducing the browsing pressure on desirable plants. Our study demonstrates that the chemical defences from one plant can be used to effectively protect another from browsing by multiple herbivore species, and that this can be achieved using a waste product of the forestry industry. Importantly, we found contrasting effects for red deer and moose when only alternate seedlings were treated, with implications for applying the extract in the field.

For red deer, the medium inter-seedling distances used in forestry resulted in a greater probability of browsing than the large and small inter-seedling distances, but this appears to have been driven by a high level of browsing of untreated seedlings with treated neighbours. Here, alternate extract application resulted in associational susceptibility for the untreated seedlings, such that untreated seedlings were more likely to be eaten when they had treated rather than untreated neighbours. This is consistent with associational susceptibility via the neighbour contrast mechanism (Atsatt and O'Dowd, 1976) and has been documented in another

Table 1
Planned comparisons testing treatment effects on the probability of Scots pine seedlings (*Pinus sylvestris*) being browsed by red deer *Cervus elaphus* (binomial GLMM with trial number as a random effect). Trials were conducted at Tovetorp Research Station, south-central Sweden, from January through March 2016, using an extract of birch bark (*Betula* spp.) applied to seedlings. Seedlings: Untreated = no extract applied, Treated = extract applied; extract application scheme: None = no repellent applied, Alt. = alternate seedlings treated with repellent, All = all seedlings treated with repellent; inter-seedling distance: Large = 5.30 m, Medium = 1.77 m, Small = 0.59 m. The estimates are given on the logit scale and P-values for comparisons significant at $\alpha = 0.05$ are given in bold.

Fixed effect		Estimate	SE	z	P	95% CI
(Intercept)		−1.116	0.158	−7.085	<0.001	−1.453 to −0.813
Untreated - Treated		0.539	0.141	3.815	**<0.001**	0.263–0.833
(Large + Small) - Medium		−0.220	0.107	−2.051	**0.040**	−0.438 to −0.002
Large - Small		0.148	0.192	0.771	0.441	−0.238 to 0.541
Large	Untreated(Alt.) - Untreated(None)	−0.091	0.220	−0.416	0.677	−0.538 to 0.347
	Treated(Alt.) - Treated(All)	−0.018	0.252	−0.072	0.943	−0.538 to 0.479
Medium	Untreated(Alt.) - Untreated(None)	0.568	0.224	2.541	**0.011**	0.131–1.030
	Treated(Alt.) - Treated(All)	0.266	0.227	1.174	0.240	−0.184 to 0.729
Small	Untreated(Alt.) - Untreated(None)	−0.308	0.238	−1.293	0.196	−0.805 to 0.158
	Treated(Alt.) - Treated(All)	−0.443	0.314	−1.412	0.158	−1.143 to 0.138

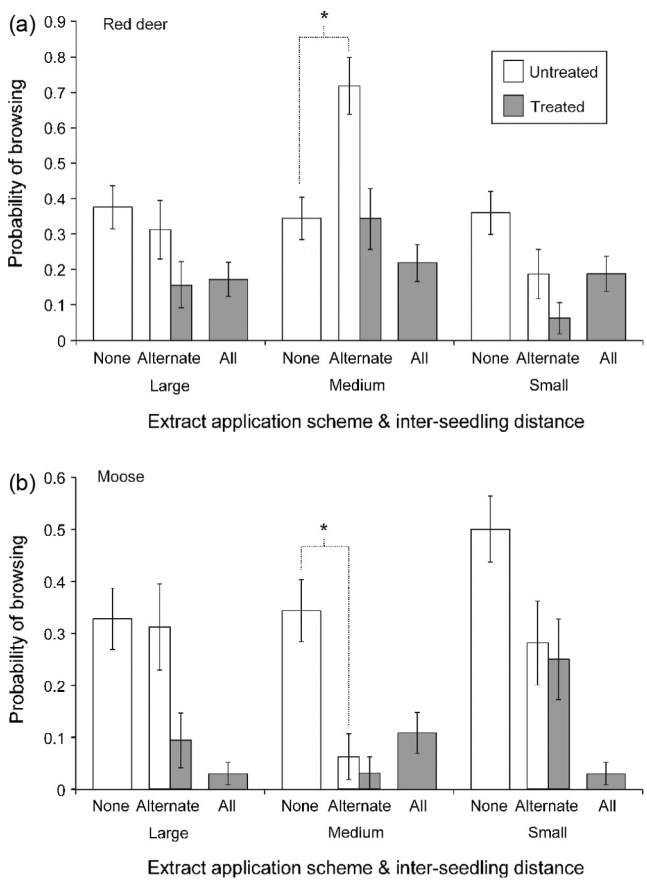

Fig. 2. Probability of untreated and treated Scots pine seedlings (*Pinus sylvestris*) being browsed by (a) red deer *Cervus elaphus* and (b) moose *Alces alces*, as a factor of inter-seedling distance and application scheme of birch bark extract (*Betula* spp.) in the patch (mean ± SE). Trials were conducted at (a) Tovetorp Research Station and (b) Öster Malma Wildlife Park, south-central Sweden, from January through March 2016. At each inter-seedling distance, seedlings were presented in one of three extract application schemes: (1) none applied, (2) applied to alternate seedlings, and (3) applied to all seedlings. Large, medium and small iner-plant distances corresponded to 5.27 m, 1.76 m and 0.59 m, respectively.

ungulate species, the fallow deer (Bergvall et al., 2006). In the latter, fallow deer ate more from a bucket of low-tannin (less defended) pellets when it was presented among seven high-tannin (more defended) buckets than among seven low-tannin buckets. We found support for this behaviour in red deer despite presenting a much higher ratio of undefended to defended food (1:1 compared to 1:7). In contrast to Bergvall et al. (2006), we found no significant difference in the probability of a treated seedling being browsed whether neighbours were untreated or treated. We therefore suggest that red deer were strongly selective between pine seedlings at inter-seedling distances typically used in forestry, with a strategy to avoid seedlings treated with birch bark extract. As a result, application of birch bark extract to alternate seedlings would not likely be an effective strategy for reducing the browsing impact of this herbivore species.

Inter-seedling distance did not affect the probability of a seedling being browsed by moose overall. However, at medium inter-seedling distances, untreated seedlings with treated neighbours were less likely to be browsed than those with untreated neighbours, corresponding to associational refuge via the repellent plant mechanism. The response of moose to alternately treated seedlings is consistent with the reduced consumption of tree seedlings in unpalatable neighbourhoods seen in other mammalian herbivores (Pietrzykowski et al., 2003; Smit et al., 2006). As in the red deer, moose did not demonstrate any significant differences in the probability of browsing treated seedlings whether they had untreated or treated neighbours. Treating alternate plants in a plantation may be a viable option for reducing browsing by moose but not by red deer; it is therefore essential to understand the herbivore assemblage of the area requiring protection.

While any amount of damage to a pine seedling may cause apical branching and reduce its economic value (Nilsson et al., 2016), the amount of foliage removed if it is browsed is critical to its ability to recover (Rosenthal and Kotanen, 1994). Red deer consumed a significantly greater percentage of foliage from pine seedlings when they were untreated than treated. Moose showed a similar pattern but the difference between the treatments was not significant. This appears to have resulted from the greater variation in foliage removed from treated seedlings by moose; the sample size for treated seedlings browsed by moose was less than half that for red deer. Interestingly, red deer consumed only twice the number of untreated as treated seedlings, while moose browsed four times the number of untreated as treated seedlings. The coarser scale of selection used by moose therefore resulted in consumption of proportionally more untreated seedlings. We suggest that maximising consumption of palatable (undefended) plants relative to unpalatable (defended) plants is more important for herbivores that consume large volumes of browse plants (e.g. moose) than those that consume more grasses (e.g. red deer), as plant quality is generally more variable in browse plants.

Our study suggests that all Scots pine seedlings should be treated with birch bark extract to reduce the probability and severity of browsing if both red deer and moose are present. However, we focused on the effects of treating seedlings at a relatively small

Table 2
Planned comparisons testing treatment effects on the probability of Scots pine seedlings (*Pinus sylvestris*) being browsed by moose *Alces alces* (binomial GLMM with trial number as a random effect). Trials were conducted at Öster Malma Wildlife Park, south-central Sweden, from January through March 2016, using an extract of birch bark (*Betula* spp.) applied to seedlings. Seedlings: Untreated = no extract applied, Treated = extract applied; extract application scheme: None = no repellent applied, Alt. = alternate seedlings treated with repellent, All = all seedlings treated with repellent; inter-seedling distance: Large = 5.30 m, Medium = 1.77 m, Small = 0.59 m. The estimates are given on the logit scale and P-values for comparisons significant at α = 0.05 are given in bold.

Fixed effect		Estimate	SE	z	P	95% CI
(Intercept)		−2.536	0.448	−5.658	<0.001	−3.600 to −1.742
Untreated - Treated		1.254	0.353	3.552	**<0.001**	0.581–2.039
(Large + Small) - Medium		0.203	0.290	0.701	0.483	−0.384 to 0.824
Large - Small		−0.237	0.478	−0.497	0.619	−1.243 to 0.752
Large	Untreated(Alt.) - Untreated(None)	0.138	0.490	0.282	0.778	−0.849 to 1.190
	Treated(Alt.) - Treated(All)	0.544	0.567	0.959	0.338	−0.568 to 1.768
Medium	Untreated(Alt.) - Untreated(None)	−1.186	0.607	−1.956	**0.050**	−2.569 to −0.059
	Treated(Alt.) - Treated(All)	−0.531	0.692	−0.767	0.443	−2.087 to 0.775
Small	Untreated(Alt.) - Untreated(None.)	−0.408	0.513	−0.796	0.426	−1.490 to 0.639
	Treated(Alt.) - Treated(All)	0.721	0.545	1.324	0.186	−0.376 to 1.874

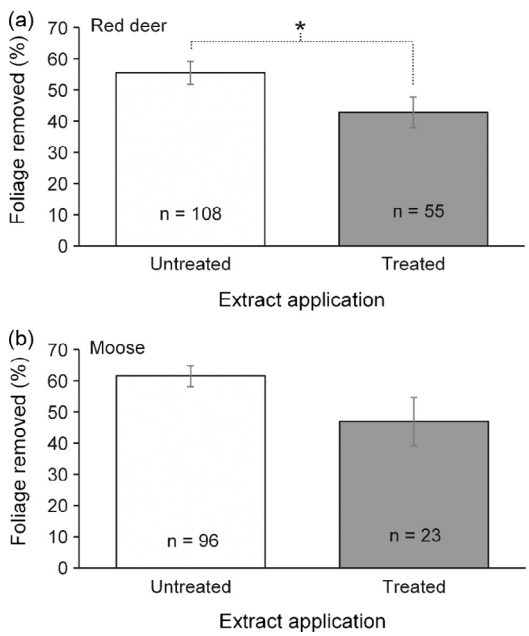

Fig. 3. Percentage of foliage removed by (a) red deer *Cervus elaphus* and (b) moose *Alces alces* from *Pinus sylvestris* seedlings that were untreated or treated with birch bark extract (mean ± SE).

scale. It is important to consider how broad-scale application of birch bark extract in a plantation or forest may alter its effectiveness to deter herbivores. In such cases, avoiding treated seedlings is much more difficult given that herbivores must achieve some minimum level of intake and searching for food is costly. Some studies have reported reduced effectiveness of particular repellents against mammalian herbivores when applied over larger areas compared to small plots, citing a lack of alternative food nearby (Conover, 1984; Gillingham et al., 1987); this is supported by research showing that flavour aversions are stronger in the presence of alternative foods (Kimball et al., 2002). These effects may be particularly important when the availability of other browse resources is low, and herbivores face a higher cost of travelling to find more palatable food (Charnov, 1976). For example, moose are known to switch to lower browse quality as winter progresses and the higher quality resources are depleted (Van Beest et al., 2010). Treating all pine seedlings is likely to be less problematic in mixed species stands or where there is alternative browse nearby.

The availability of alternative foods not only affects choices between plants and plant patches directly, but it also determines the herbivore's nutritional state and this, in turn, influences future decisions. We tested captive animals receiving a supplemented diet and so they were probably more nutritionally satisfied than their wild counterparts. As a result, captive animals could afford to be more selective, and therefore may have rejected treated seedlings more often than they would under natural conditions. We argue that this effect was probably balanced by the much higher animal densities in enclosures compared to those in the wild, so pressure on the available resources, including the pine seedlings, was much higher than would typically be encountered in a forest. Field experiments with planted seedlings exposed to free-ranging red deer and moose would be the next step to test the effects of large-scale extract application on foraging behaviours. With high numbers of seedlings, this might also mean that some of the differences between treatments that were statistically non-significant in our study, possibly due to large errors, could become important.

Browsing mammals significantly impact on the successful regeneration of trees and the productivity of plantations in many parts of the world. Harnessing the plant chemical defences that have evolved over evolutionary time, associated with post-ingestive consequences, could provide an effective avenue for non-lethal control of browsing. We have demonstrated that birch bark extract deters red deer and moose from browsing Scots pine seedlings, but the idea is not confined to this system. Many plants possess anti-herbivore defences that could be used to protect less defended plant species from browsing by mammalian herbivores, particularly those with a generalist diet as they cannot evolve physiological mechanisms for dealing with the variety of defence chemicals they encounter (Jogia et al., 1989). For example, the compounds in the bark of willows (*Salix* spp.) are associated with reduced consumption by mountain hares *Lepus timidus* (Tahvanainen et al., 1985), foliar compounds in *Eucalyptus globulus* limit intake by the common brushtail possum *Trichosurus vulpecula* (O'Reilly-Wapstra et al., 2004), and foliar compounds in trees of the Miombo biome dictate selective herbivory by African elephants *Loxodonta africana* (Jachmann, 1989). The protection of palatable plants using the existing plant defensive chemistry in less palatable plants presents an exciting avenue for research into behavioural mechanisms for browsing management, particularly where this can be achieved using resources that would otherwise be wasted.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.foreco.2017.01.013.

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